$P(E^c) = 1 - P(E)$

Demorgan's: $\overline{(E \cup F)} = \overline{E} \cap \overline{F}$

$\overline{(E \cap F)} = \overline{E} \cup \overline{F}$

$\binom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$

$P(A) = P(AB) + P(A-B)$

Classical: $\dfrac{\text{desired}}{\text{total}}$

$0 \leq P(E) \leq 1$, $P(S) = 1$

disjoint: $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

conditional: new info shrinks sample space

$P(E|F)$: probability of E given F

$0 \leq P(E|F) \leq 1$, $P(S|F) = 1$

disjoint: $P((E_1 \cup E_2)|F) = \sum_n P(E_n|F)$

$$P(E|F) = \frac{P(EF)}{P(F)}$$

$$P(EF) = P(E|F)P(F) = P(F|E)P(E)$$

Partition Rule

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$P(B) = \sum P(B|A_n)P(A_n)$$

$\longrightarrow$ conditional

$$P(B|E) = P(B|A_1E)P(A_1|E) + \ldots + P(B|A_nE)P(A_n|E)$$

Chain Rule: things happening in a row

$$P(A_1 A_2 A_3) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 A_2)$$

$$\rightarrow P(A_1 A_2 \ldots A_n) = P(A_1) \cdot P(A_2 | A_1) \cdot P(A_3 | A_1 A_2) \cdots P(A_n | A_1 A_2 \cdots A_{n-1})$$

Bayes Formula:

$$P(A_i | B) = \frac{P(B | A_i) P(A_i)}{\sum_{j=1}^{n} P(B | A_j) P(A_j)}$$

flipped: $P(B|A_i) \longleftrightarrow P(A_i | B)$

$$(\text{prob of cause } A) = \frac{\text{prob of result given cause } A}{\text{total prob of result from all causes}}$$

Events E, F are <u>independent</u> if $P(EF) = P(E)P(F)$

$$P(F) \neq 0 \longrightarrow P(E|F) = P(E)$$

$\rightarrow$ statistical independence: information, not physical interaction

E is ind. of F $\rightarrow$ learning that F has occurred gives no new info to help predict if E will occur

if E, F are ind., then $\longleftrightarrow$ 1) E, F^c are ind.

E^c, F are ind.

E^c, F^c are ind.

(if knowing F happened doesn't help predict E, then knowing F didn't happen doesn't help predict E either)

$$P(EF^c) = P(E)P(F^c|E)$$

disjoint events are not independent

independence of variables implies independence of events

Events A, B, C are independent if

1) $A \perp\!\!\!\perp B$
2) $B \perp\!\!\!\perp C$
3) $A \perp\!\!\!\perp C$
4) $P(ABC) = P(A)P(B)P(C)$

$P(ABC) = P(A)P(B)P(C)$ does not imply pairwise independence

Conditional Independence

$A \perp\!\!\!\perp B \mid C$ (conditionally independent given C) if

$P(AB|C) = P(A|C)P(B|C)$ assuming $P(C) \neq 0$

- if we know C has happened, learning B gives no info on A

$$A \perp\!\!\!\perp B \mid C \longleftrightarrow P(A|BC) = P(A|C)$$

$$\longrightarrow P(A-B) = P(A)(1-P(B))$$

$P(AB) < P(A)P(B) \rightarrow$ repulsion ($P(AB) = 0 \rightarrow$ disjoint)

$P(AB) = P(A)P(B) \rightarrow$ independence

$P(AB) < P(A)P(B) \rightarrow$ attraction

geometric series ; $\sum_{k=0}^{\infty} ar^k \frac{a}{1-r}, |r|<1$ sum $= \frac{\text{1st term}}{1 - \text{ratio}}$

$$\sum_{k=0}^{n-1} ar^k = a\frac{1-r^n}{1-r}, \quad r \neq 1$$

$$\sum_{k=1}^{n} ar^{k-1} = a\frac{1-r^n}{1-r}$$

Independent trials

P of getting K occurences of E in n trials:

$$P(\text{exactly } k \text{ successes}) = \binom{n}{k} P(E)^k (1 - P(E))^{n-k}$$

$\binom{n}{k} = \binom{n}{n-k}$ if rolling 5 dice, choosing 2 as success is the same as choosing 3 as failures

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = 1 \quad \text{(binomial theorem)}$$

- if you add up the probability of getting 0, 1, 2, ... n successes, it must total to 1 binomial distribution function

- probability of getting k successes in n independent trials, where each trial has the same probability of success p

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Random Variables

function that maps outcomes in a sample space to a real number

$$X : S \rightarrow R$$

Let $A \subseteq \mathbb{R}$ event

$$\{X \in A\} = \{u \in S : X(u) \in A\}$$

u : a specific outcome

X(u) : scalar assigned to that outcome

CDF

CDF : cumulative distribution function

$$F_X(u) = P(X \leq u) \quad \text{for} \quad -\infty < u < \infty$$

which random var.

- tells accumulated probability $P(X \leq u)$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

General : $F_X(u) = \sum_{k \le u} P(X=k)$

Geometric Series : $F_X(u) = \sum_{k=1}^{\lfloor u \rfloor} (1-p)^{k-1} p$

$\rightarrow F_N(u) = 1-(1-p)^{\lfloor u \rfloor}$

CDF properties

- $0 \le F_X(u) \le 1$
  - $\rightarrow$ CDF outputs P, so value $\in [0,1]$
- if $a < b$, then $F(a) \le F(b)$

  since: $P(X \le b) = P(X \le a) + \underbrace{P(a < X \le b)}_{\ge 0}$

  $\rightarrow$ monotone non decreasing

  if $a < b$, then $F(a) \le F(b)$
- $F(\infty) = \lim_{u \to \infty} F(u) = 1$

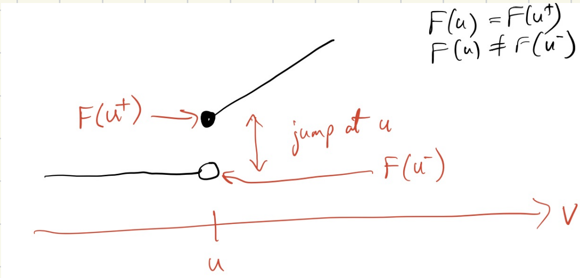
  $\rightarrow P(X) < \infty = 1 = 100\%$

  (all possible outcomes $< \infty$)
- $F(-\infty) = \lim_{u \to -\infty} F(u) = 0$

  $\rightarrow P(X) < -\infty = 0 = 0\%$

  (no outcomes $< -\infty$)

- $F(u)$ is right-continuous



- $P(X>u) = P(\{X \le u\}^c) = 1 - P(X \le u) = 1 - F(u)$
- $P(X<u) = F(u^-)$
- $P(X \ge u) = 1 - P(X<u) = 1 - F(u^-)$
- $P(X=u) = \underset{\lim R}{F(u)} - \underset{\lim L}{F(u^-)}$

$P(X>u) = 1 - P(X \le u)$

$\longrightarrow P(X=u) =$ height of jump @ u

- $P(a<X \le b) = P(X \le b) - P(X \le a)$

$= F(b) - F(a)$

- $P(a<X<b) = F(b^-) - F(a)$
- $P(a \le X < b) = F(b^-) - F(a^-)$
- $P(a \le X \le b) = F(b) - F(a^-)$

If $F(u)$ cont. @ $u=a$, then no jump @ a

$\longrightarrow P(a<X \le b) = P(a \le X \le b)$

since $F(a) = F(a^+) = F(a^-)$ (lines meet perfectly)

$\therefore P(X=a) = 0$

Discrete Random Variable

→ can only take finite (or countably infinite) set of distinct values

→ CDF graph is staircase function

→ $P(x=a) > 0$

Continuous Random Variable

→ can take any real number within a given range

→ CDF graph is continuous (no jumps)

→ $P(x=a) = 0$, only measure P in an interval

→ differentiable, exept possibly at a finite or countably finite set of points

→ monotone non-decreasing, derivative always non-negative ($F_x(u)$ is cumulative, P cannot go down)

Probability Mass Function (PMF)

pmf of a discrete r.v. X:

$$P_x(u) = P(X=u)$$

p (pmf) P (probability)

→ "lookup table" that maps how much P belongs to each distinct outcome

→ is plugging a number u into the PMF, it tells us the P that $X = u$

- $p_x(u) \geq 0 \ \forall u$

  → P of any specific outcome ≥ 0

- $\sum_u p_x(u) = 1$

  → sum of p of all outcomes $= 1$

- Uniform Discrete r.v

  → pmf has finite # of equally spaced and equal height values

  → e.g. dice roll

- Binomial r.v.

$$p_x(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

for $k = 0, 1, 2, \ldots, n$

→ counts # of successes in fixed # of trials

$$\rightarrow 1 = \sum_{k=0}^{n} p_x(k) = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

- Poisson r.v. (with parameter λ)

$$p_x(k) = \frac{\lambda^k e^{-\lambda}}{k!} \quad \text{for } k = 0, 1, 2, \ldots n$$

→ counts how many times an event happens over a fixed amount of time or space

→ e.g. how many texts will I get in the next hour?

→ λ represents average rate at which things happen (4 texts per hour → $\lambda = 4$)

→ $$\sum_{k=0}^{\infty} p_X(k) = \sum_{k=0}^{\infty} \frac{\lambda^k e^{-\lambda}}{k!}$$

- Geometric r.v. (parameter

$$p_X(k) = p(1-p)^{k-1} \text{ for } k = 1, 2, 3 \ldots$$

→ how many trials (X) did it take to get a success?

→ parameter p is P of success on a single try

$$P(X > k) = \sum_{j=k+1}^{\infty} p_X(j) = (1-p)^k$$

CDF: $F_X(u) = P(X \leq u) = 1 - P(X > u) = 1 - (1-p)^u$

Probability Density Function (pdf)

pdf of a cont. r.v. X is:

$$f_X(u) = \frac{d}{du} F_X(u) \qquad f \to \text{pdf} \qquad F \to \text{CDF}$$

(derivative of CDF) (density) (cumulative)

→ continuous version of pmf which is for discrete

- $\int_a^b f_X(u)\,du = F_X(b) - F_X(a) = P[a \leq X \leq b]$

  → to find $P(a \leq X \leq b)$, find the area under the PDF curve between those two points

  → $P(X < a) = P(X \leq a)$, since $P(X = a) = 0$

- $f_X(u) \geq 0$

  → bc CDF is monotone non decreasing

  → pdf value can be > 1, since value is not p, area is

- $\int_{-\infty}^{\infty} f_X(u)\,du = P(-\infty < X < \infty) = 1$

- recover CDF from pdf:

$$F_X(u) = \int_{-\infty}^{u} f_X(v)\,dv$$

- $A \subseteq \mathbb{R}$

$$P[X \in A] = \int_A f_X(u)\,du$$

* when pdf limited to a window, (e.g. $a < x < b$), CDF at least 3 pieces
0 before, $\int$ inside, 1 after

- Uniformly distributed r.v. on an interval (a,b) has [a,b] pdf

$$f_X(u) = \begin{cases} \frac{1}{b-a}, & a < u < b \\ 0, & \text{else} \end{cases}$$

Gaussian (normal) r.v. X has pdf

$$f_X(u) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{u-m}{\sigma}\right)^2} \quad \text{for } -\infty < u < \infty$$

m = mean

σ^2 = variance

σ = std dev

$X \sim N(m, \sigma^2)$

$\sim$ is distributed as

N normal

- standard/unit Gaussian

$m=0, \sigma^2=1 \qquad X \sim N(0,1)$

Expectation (expected value, avg, mean)

if X is a discrete r.v., then its expected value (or mean) is:

$$E[X] = \sum_u u \cdot P_X(u)$$

for continuous r.v.:

$$E[X] = \int_{-\infty}^{\infty} u f_X(u)\, du \longleftrightarrow E[g(x)] = \int g(x) f(x)\, dx$$

- Uniform Distribution (continuous or discrete)

$$X \sim U(a,b)$$

$$E[X] = \frac{a+b}{2}$$

- Binomial Distribution

$$X \sim Bin(n,p)$$

n = trials
p = p of success

$$E[X] = np$$

- Poisson Distribution

$$X \sim Pois(\lambda)$$

$$E[X] = \lambda$$

- Bernoulli Distribution (Bin n=1)

$$X \sim Bin(1,p)$$

$$E[X] = p$$

- Geometric Distribution

$$X \sim Geo(p)$$

p = p of success on a single trial

$$E[X] = \frac{1}{p}$$

- Gaussian (Normal) Distribution

$$X \sim N(m, \sigma^2)$$

$\mu \leftrightarrow m$

$$E[X] = m$$

- Exponential Distribution

$$X \sim Exp(\lambda)$$

$$E[X] = \frac{1}{\lambda}$$

Shortcuts

- EV of a step function pdf $\quad A(\text{Area}) = (b-a)h$

$M(\text{Midpt}) = \frac{a+b}{2}$

$$E[X] = \sum_{i=1}^{n} (A_i \cdot M_i)$$

- for a r.v. bounded between a and b:

$$E[X] = \int_0^{\infty} (1 - F_X(u))\,du - \int_{-\infty}^{0} F_X(u)\,du$$